MEDICINAL USE OF CARBOHYDRATES







Lecture Objectives

At the end of this session student will be able to understand;

1. The importance of carbohydrates for life.

2. Interaction of carbohydrates in biological process.

3. Biochemical and medicinal importance of carbohydrates.

Importance of Carbohydrates

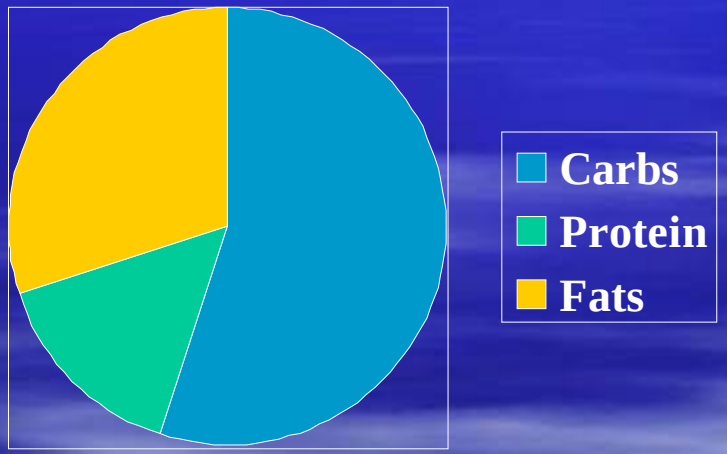
Can you live without sugar?

- Soda/Punch
- Cookies
- Candy
- Chocolate
- Desserts
- Sugary Cereals
- Ice cream



CARBOHYDRATES

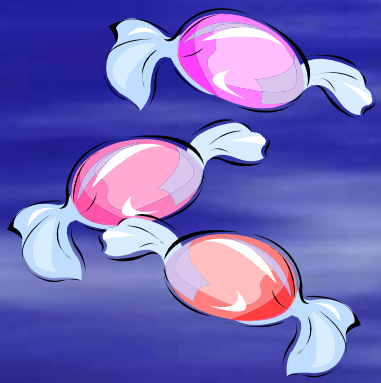
1. **<u>60%</u> of our food should come from carbohydrates.**



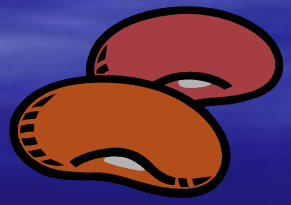
2. Carbohydrates are grains, fruits, vegetables, legumes and sugar.











Carbohydrates

- ***GLUCOSE*** provides energy for the brain and ½ of energy for muscles and tissues
- ***GLYCOGEN*** is stored glucose
- ***GLUCOSE*** is immediate energy
- ***GLYCOGEN*** is reserve energy

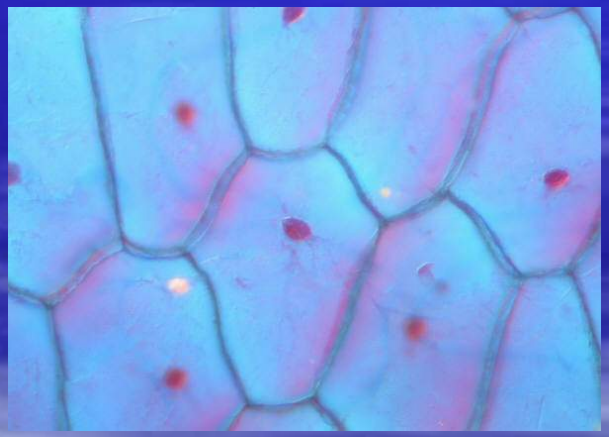
CARBOHYDRATES

3. Carbohydrates give the body energy. They are the best source of fuel for the body. Carbohydrates also help to digest protein and fat.



CARBOHYDRATES

4. If we eat more carbohydrates than are needed for energy, the extra is stored in the liver or in the tissues as fat.



Carbohydrates

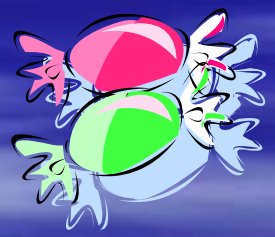
- 5. all plant food
- Milk (LACTOSE)

- carbohydrates are not equal
  - simple carbohydrates
  - complex carbohydrates

Simple Carbohydrates

6. Simple carbohydrates are quick energy sources. They come from sugar. They do not usually supply any other nutrients or fiber.



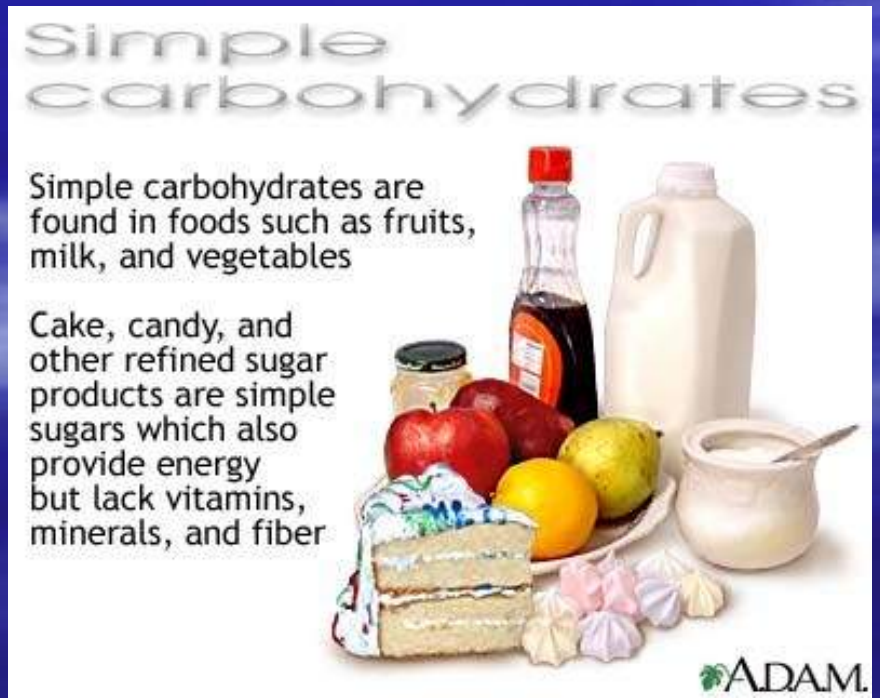






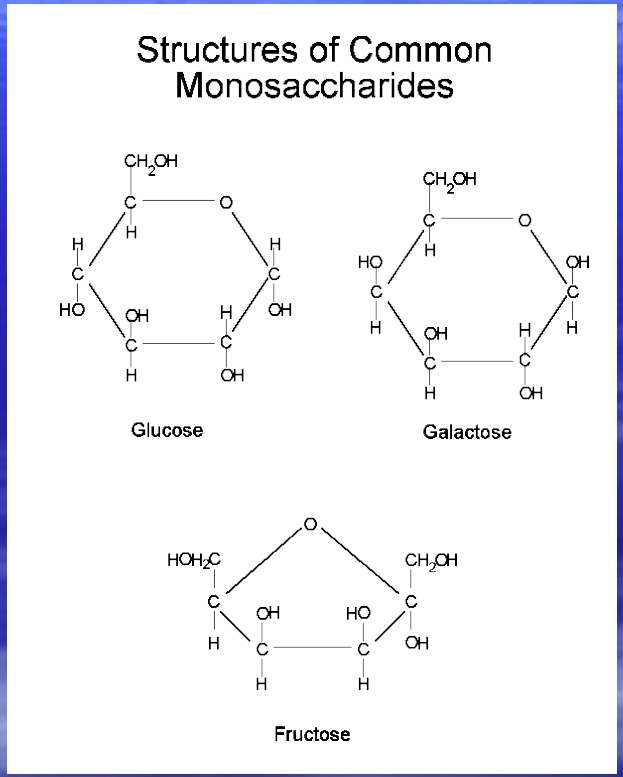
Simple Carbohydrates

- sugars
  - monosaccharides – single sugars
  - disaccharides – 2 monosaccharides



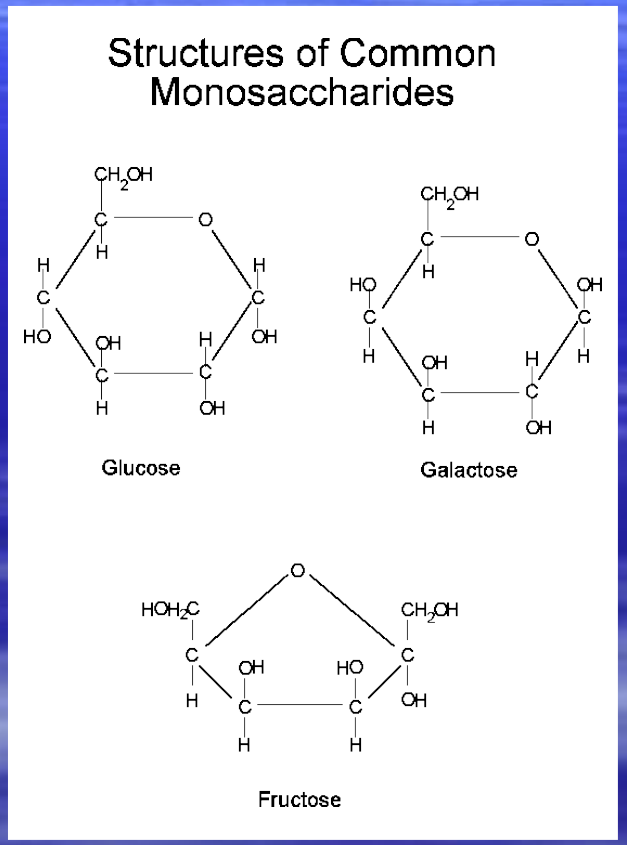
Glucose

- mild sweet flavor
- known as blood sugar
- essential energy source
- found in every disaccharide and polysaccharide



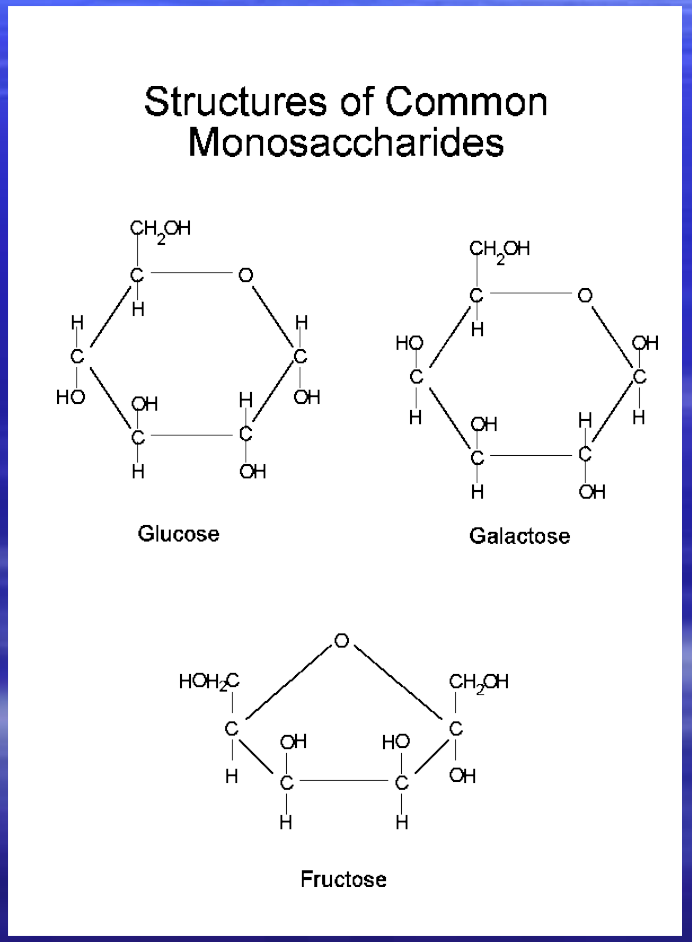
Fructose

- sweetest sugar
- found in fruits and honey
- added to soft drinks, cereals, deserts



Galactose

- hardly tastes sweet
- rarely found naturally as a single sugar

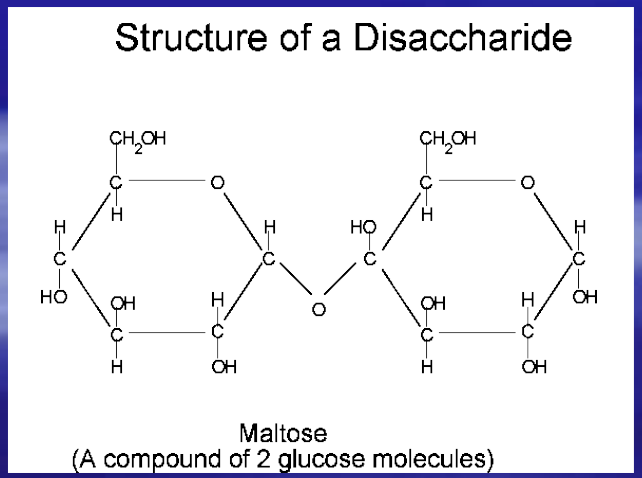


Disaccharides

- pairs of the monosaccharides
  - glucose is always present
  - 2nd of the pair could be fructose, galactose or another glucose
  - taken apart by hydrolysis
  - put together by condensation
  - hydrolysis and condensation occur with all energy nutrients
  - maltose, sucrose, lactose

Maltose

- 2 glucose units
- produced when starch breaks down
- not abundant





Sucrose

- fructose and glucose
- tastes sweet
  - fruit, vegetables, grains
- table sugar is refined sugarcane and sugar beets
- brown, white, powdered







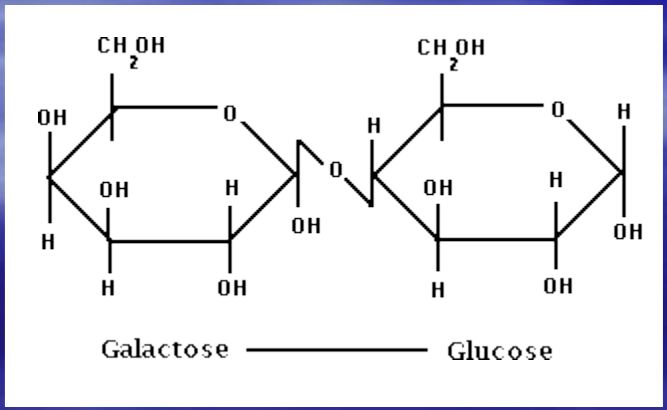




Lactose

- glucose and galactose
- main carbohydrate in milk
  - known as milk sugar





Carbohydrates & Biological processes.

Simple Sugars



7. Glucose or blood sugar is the basic source of energy for all living things.
8. Sucrose or table sugar is made from sugar beets or sugar cane.
9. Fructose is sugar found in fruit, honey and vegetables.
11. Maltose is grain starch broken down into sugar.
12. Lactose is milk sugar.

Complex Carbohydrates

- starches and fibers
- polysaccharides
  - chains of monosaccharides



Starches

- stored in plant cells
- body hydrolyzes plant starch to glucose.
- All starch bases food products are from plant origin, seeds are the richest source; 70% of their weight is starch



STARCHES

- 14. Starch in the body breaks down simple sugars. The body has to break down all sugar and starch into glucose to use it.





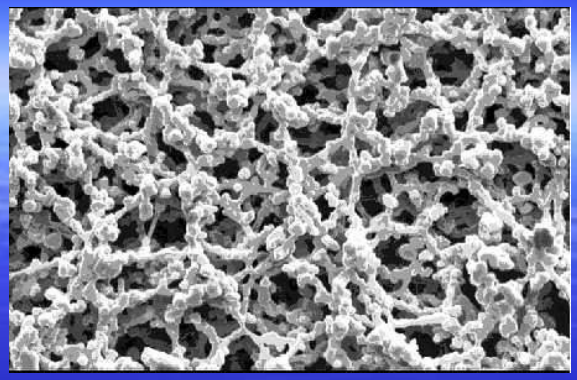
STARCH	GLYCOGEN	CELLULOSE
- Storage carbohydrate in plants - Polymer of α D-glucose - Comprises of Amylose and Amylopectin 1. Amylose- long straight chain consisting of glucose linked by α (1-4) glycosidic linkages. 2. Amylopectin- has branching at approximately every 24 glucosyl residues, linked by α(1-6) glycosidic linkages	- Storage carbohydrate in animals - **Polymer of α D-glucose** - Similar in structure to amylopectin, but approximately every 10 glucosyl residues are branched.	- Provide structural support in plant cell walls. - Polymer of β D-glucose. - Consist of long linear chains linked by β (1-4) linkages. - When packed in layers it makes fibrils which have high tensile strength.

Fiber

- structural parts of plants
  - found in all plant derived food
- bonds of fibers cannot be broken down during the digestive process
  - minimal or no energy available

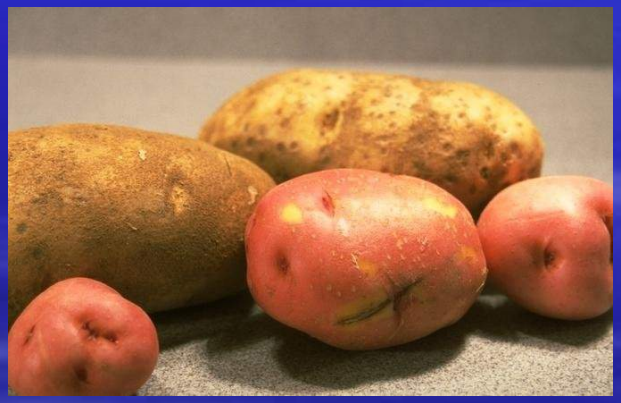


Fiber types





- cellulose
- pectins
- lignins
- resistant starches
  - classified as fibers
  - escape digestion and absorption





Fiber

1. The average American does not get enough **FIBER** in their diets.
2. The National Cancer Institute recommends that the average person gets **20-35 GRAMS** of fiber every day.
3. Two other common names for fiber are: **ROUGHAGE** or **CELLULOSE.**

4. Fiber is important because it attracts **WATER** to the **INTESTINES** and helps move food through our systems faster. You have to have water along with fiber or it is not as effective.
5. Benefits of fiber include a lowered risk of **DIVERTICULITUS**, **HEMORRHOIDS** and **COLON** or **RECAL CANCER**.

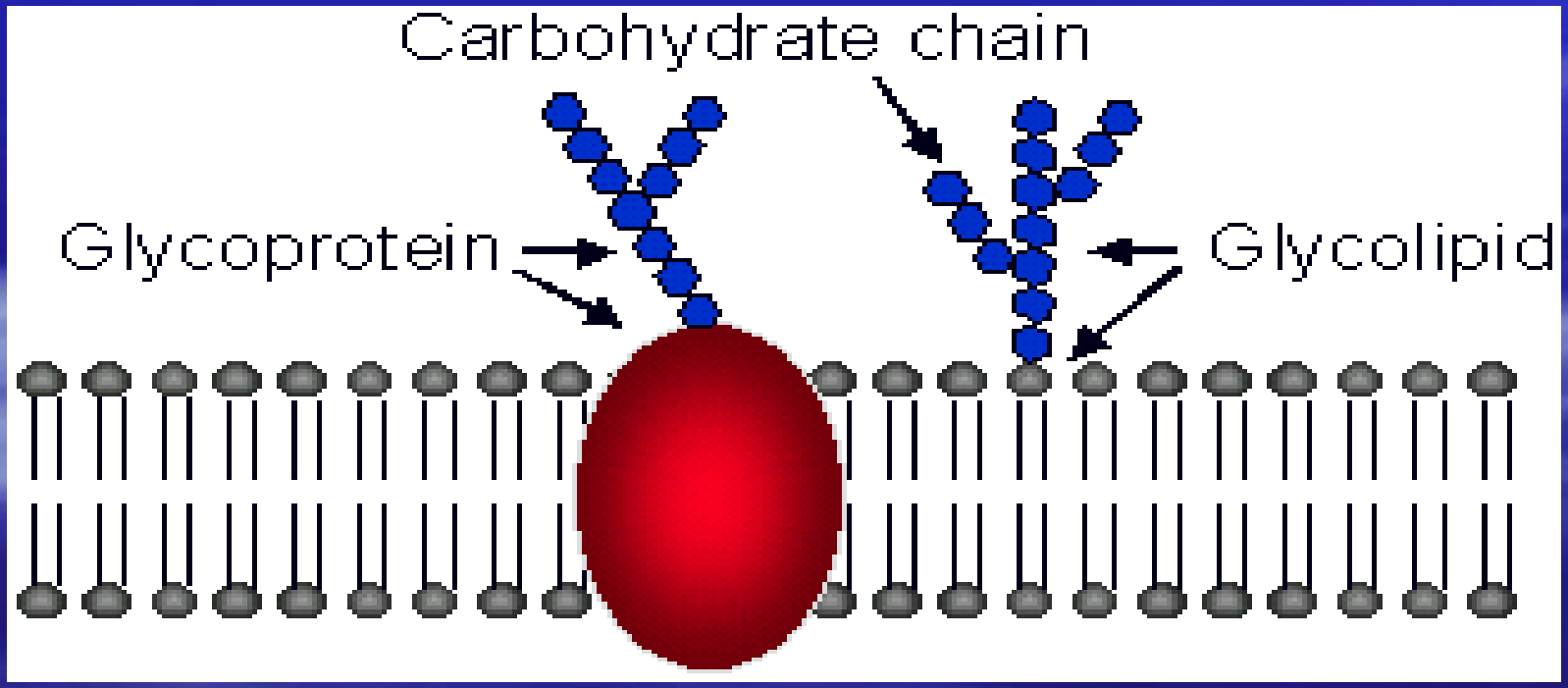
Biochemical and Medicinal Importance of Carbohydrates.

Mucopolysaccharides	Composition	Importance
1. Hyaluronic acid	1. *D*-Glucuronic acid 2. *N*-Acetyl-*D*-glucosamine	Occurs in synovial fluid, skin and vitreous humour.
2. Chondroitin sulphate	1. *D*-Glucouronic acid 2. *N*-Acetyl-*D*-galactosamine sulphate	Components of cartilage, tendons and skin (connective tissue).
3. Dermatan sulphate	1. *L*-Iduronic acid 2. *N*-Acetyl-*D*-galactosamine sulphate	Present in skin
4. Keratan sulphate	1. *D*-galactose 2. *N*-Acetyl *D*-glucosamine sulphate	Components of cartilage and cornea
5. Heparin	1. *L*-Ioduronic acid (sulphated) 2. *N*-Acetyl *D*-glucosamine (sulphated)	Present in most cells especially in liver, lungs and arterial walls.

Type	Component(s)	Examples	Physiological Effects	Major Food Sources
Insoluble (Poorly Fermented)				
Noncarbohydrate	Lignins	Wheat bran	Increases fecal bulk; estrogen-like effects	Whole grains
Carbohydrate	Cellulose	Wheat products	Increases fecal bulk	All plants
	Hemicelluloses	Brown rice	Decreases intestinal transit time	Wheat, rye, rice, vegetables
Soluble (Viscous)				
Carbohydrate	Pectins, gums, mucilages, some hemicelluloses	Apples, bananas, oranges, carrots, barley, oats, kidney beans	Delays gastric emptying; slows glucose absorption; can lower blood cholesterol	Citrus fruits, oat products (beta-glucan in particular), beans, thickeners added to foods

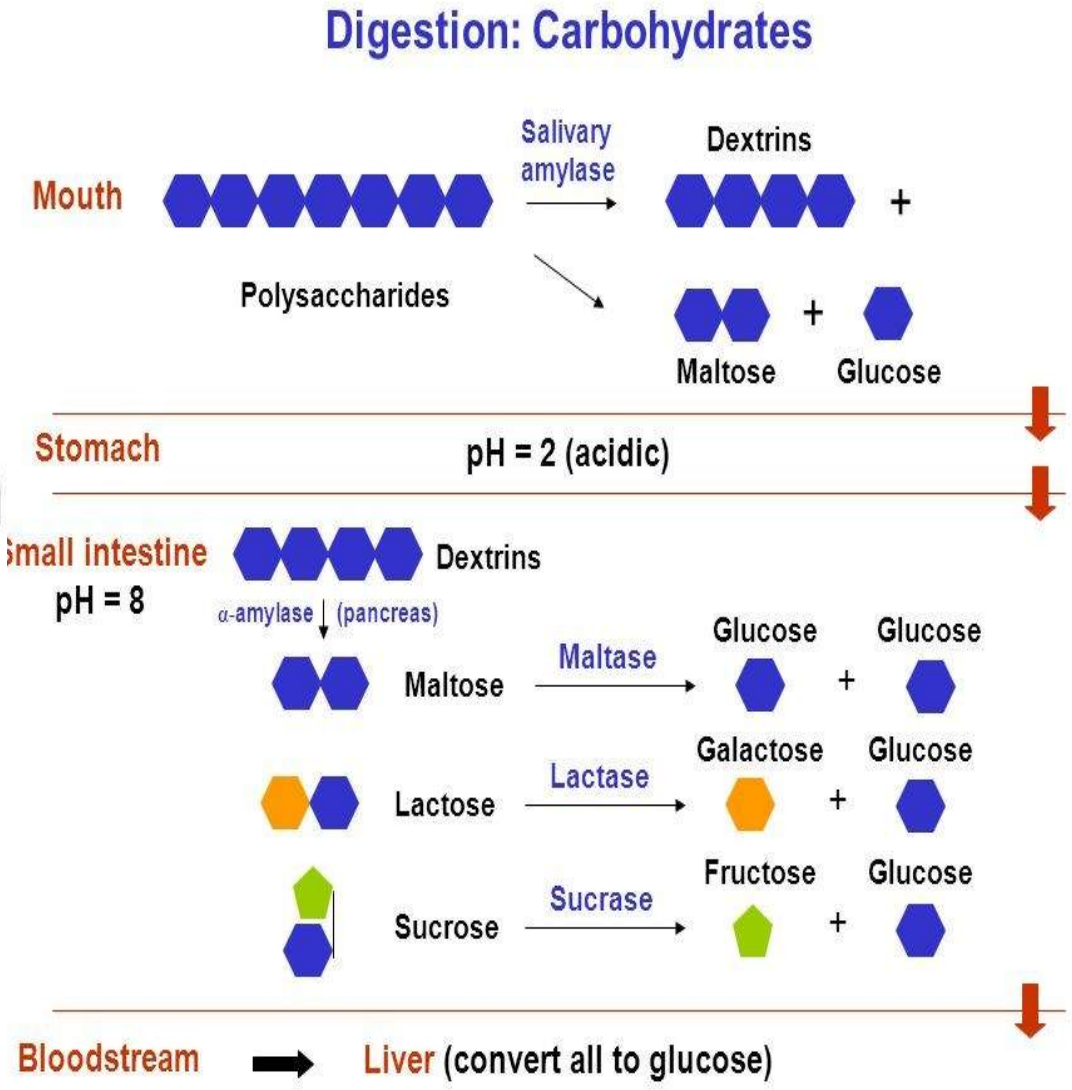
- Various food preparations, such as baby and invalid foods, are produced by hydrolysis of grains and contain large amounts of maltose, are thus easily digestible
- ***Sucrose*** **if introduced parenteral cannot be utilized,** but it can change the osmotic condition of the blood and causes a flow of water from the tissues into the blood. Thus **clinicians use it in oedema like cerebral oedema.**

- Many newly synthesized glycoproteins such as, immunoglobulin (antibodies) and peptide hormones, contain oligosaccharide carbohydrate.



Dextrin

- Produced during the hydrolysis of starch.
- An intermediate between starch and maltose.
- Used in the preparation of adhesives.
- Forms sticky colloidal suspensions with water.

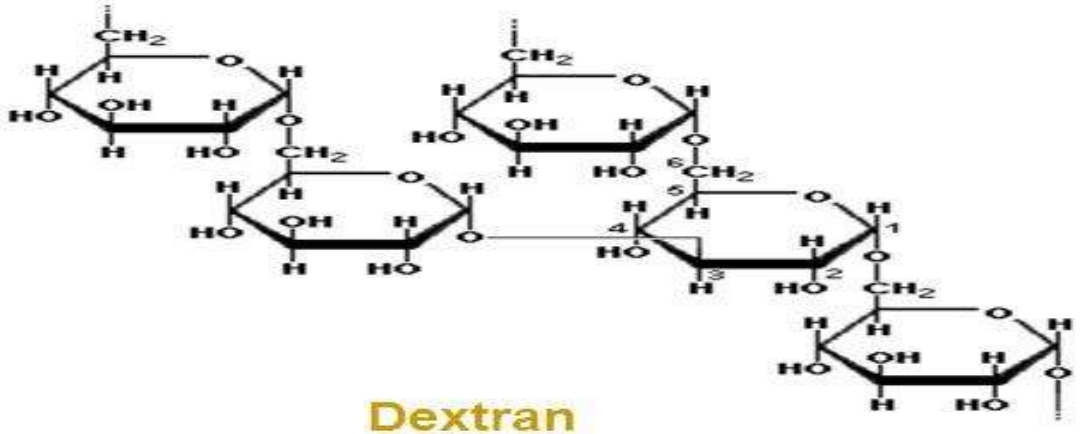


starch → dextrins → maltose → glucose

polysaccharide *polysaccharide* *disaccharide* *monosaccharide*

Dextran

- Dextran is a polysaccharide similar to amylopectin, but the main chains are formed by 1α→6 glycosidic linkages and the side branches are attached by 1α→3 or 1α→4 linkages.
- Dextran is an oral bacterial product that adheres to the teeth, creating a film called plaque.
- It is also used commercially in confections, in lacquers, as food additives, and as plasma volume expanders.



Dextran

- Produced by certain bacteria when they are grown on sucrose.
- Used as blood extenders to hold water in the bloodstream and help prevent drops in the blood volume and blood pressure.

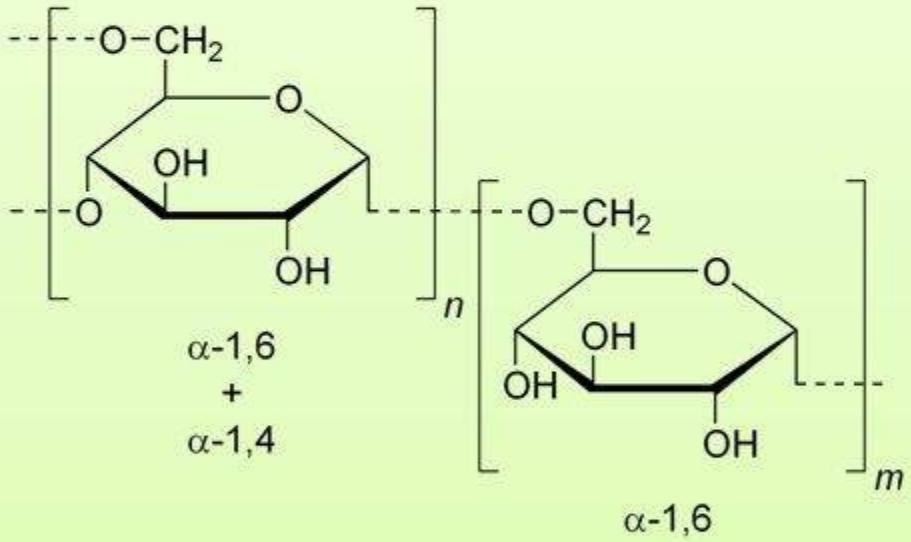
Dental plaque



- *Dental plaque is bacterial aggregations on the* teeth that cannot be removed by the strong water spray.

- Bacteria secrete a sticky matrix of an insoluble polysaccharide known as *dextran, which* allows the bacteria to stick firmly to the teeth

  Dextran is a branched polymer of the glucose that is produced by hydrolysis of sucrose from food.



- The bacterial mass (known as biofilm), their sticky matrix, and the glycoprotein film together comprise dental plaque.

C) INULIN:-

- Inulin is a polymer of fructose.
- It occurs in Dahlia bulbs, garlic, onion
- It is of low molecular weight(5000) polysaccharides , easily soluble in water

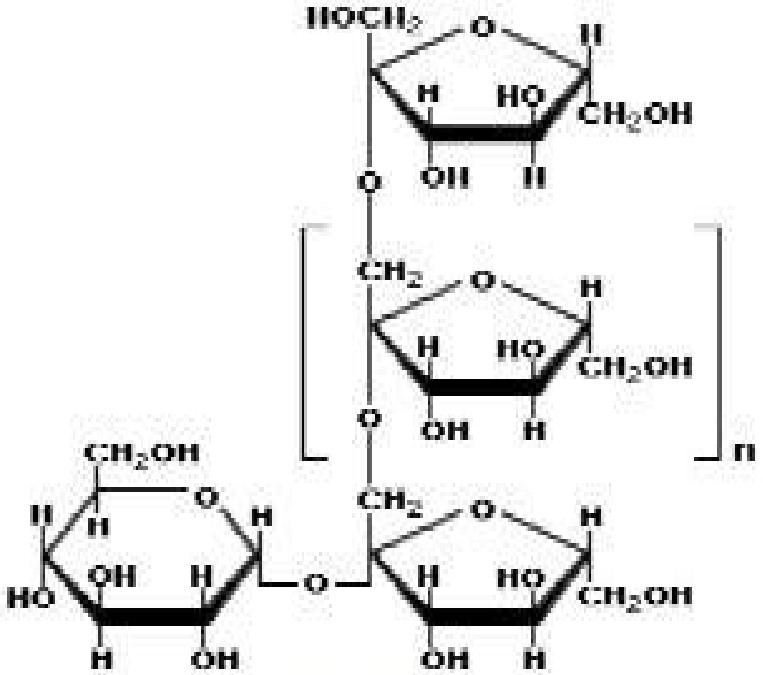


FIG 3 - INULIN

➢ Structure of Inulin:-

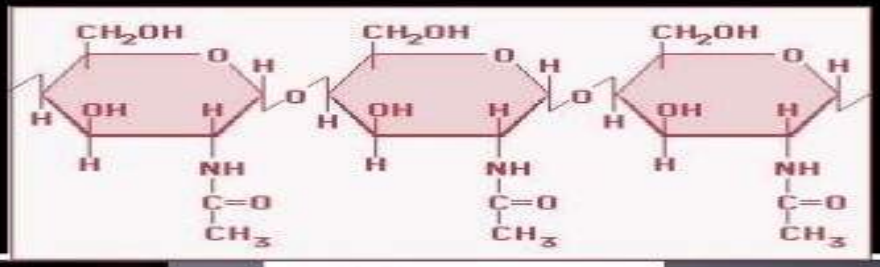
- *Inulins, also called fructans, are polymers consisting of fructose units that typically hav a terminal glucose.*
- *Oligofructose has the same structure as inulin, but the chains consist of 10 or fewer fructose units. Oligofructose has approximately 30 to 50 percent of the sweetness of sugar.*

Chitin- Structural Polysaccharide

-Chitin is the second most abundant carbohydrate polymer of N- Acetyl Glucosamine

-present in the cell wall of fungi and in the exoskeletons of crustaceans, insects and spiders

-chitin is used commercially in coatings (extends the shelf life of fruits and meats

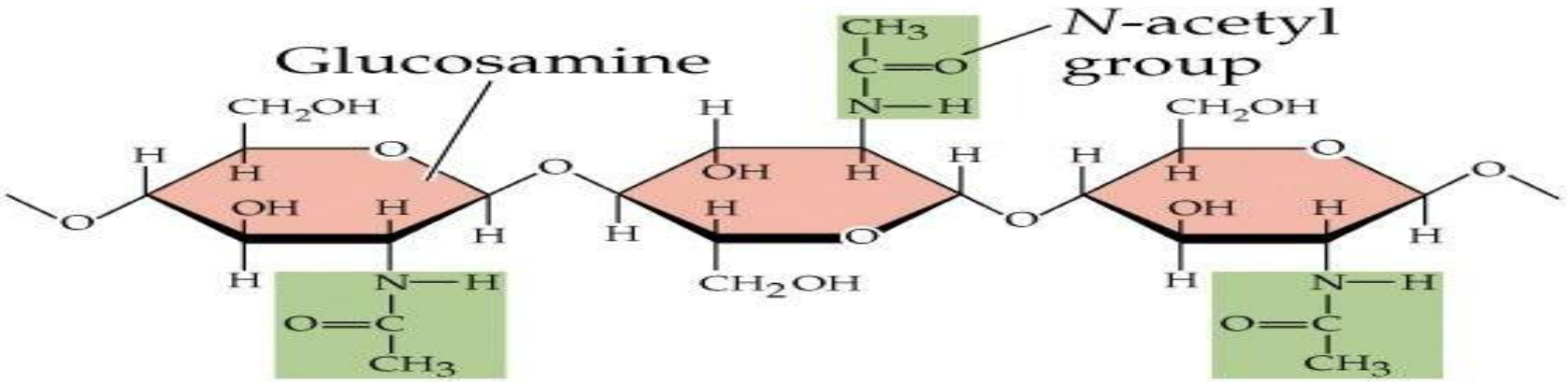








(c) **Chitin**



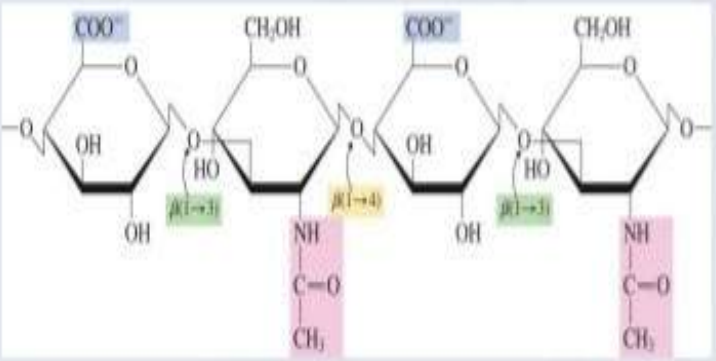
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Hyaluronic acid

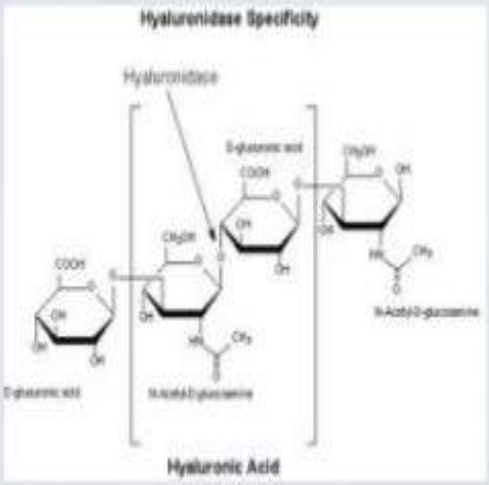
Acidic Polysaccharides

Hyaluronic acid

- repeating unit is a disaccharide composed of β-D-glucuronic acid and N-acetyl-D-glucosamine in a β-(1→3)-linkage.
- each disaccharide is attached to the next by β-(1→4)-linkage
- alternating β-(1→3) and β-(1→4)-linkages

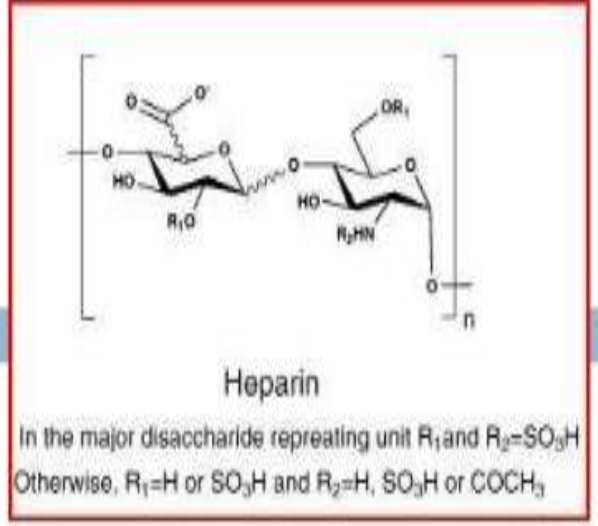


- highly viscous - serve as lubricants in the fluid of joints and part of vitreous humor of the eye
- when some insects sting, they inject an enzyme called hyaluronidase, which breaks hyaluronic acid linkages and facilitates the spread of the venom





Heparin



Heparin

In the major disaccharide repeating unit R_1 and R_2=SO_3H
Otherwise, R_1=H or SO_3H and R_2=H, SO_3H or $COCH_3$

- Due to high content of sulfo and carboxyl groups, heparin has the highest negative charge density of any known biological molecule
- It is extracted mainly from mucosal tissues of porcine and bovine
- Heparin has been used as an anticoagulant since the 1930s
- Beyond its anticoagulant activity, it shows antiviral activity and regulate angiogenesis

Heparin and Heparan Sulfate

Heparin is linear polymer, 3–40 kDa

Heparan sulfate is heparin-like polysaccharide but attached to proteins

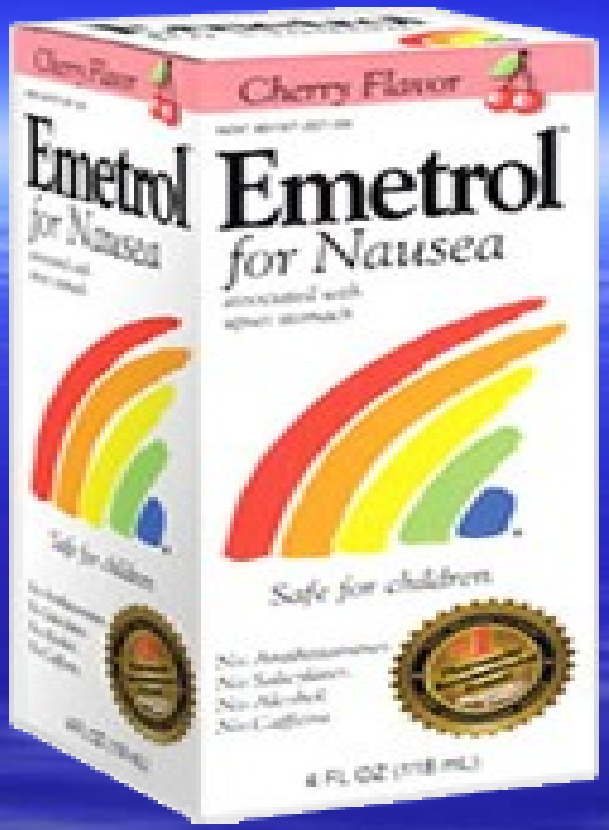
Highest negative charge density biomolecules

Prevent blood clotting by activating protease inhibitor antithrombin

Binding to various cells regulates development and formation of blood vessels

Can also bind to viruses and bacteria and decrease their virulence

Glucose/fructose/phosphoric acid (trade name Emetrol) is an over-the-counter antiemetic taken to relieve nausea and vomiting. Made by WellSpring Pharmaceutical Corporation, it was formerly distributed by McNeil Consumer Healthcare.



MANNITOL

- **Mannitol,** an alcohol derivative of simple sugar mannose, was introduced in 1960 and has since remained the major osmotic agent of choice in clinical practice.
- Its long **duration of action** (4–6 hours) and relative stability in solution have enhanced its use over the years.
- The **extra- osmotic properties of mannitol** provide additional beneficial effects in brain injury,
  1. Decreases in blood viscosity,
  2. Resulting in increases in rCBF and CPP
  3. and a resultant cerebral vasoconstriction leading to decreased CBV
  4. free radical scavenging
  5. Inhibition of apoptosis

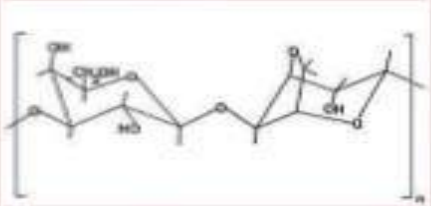


AGAR

Source:

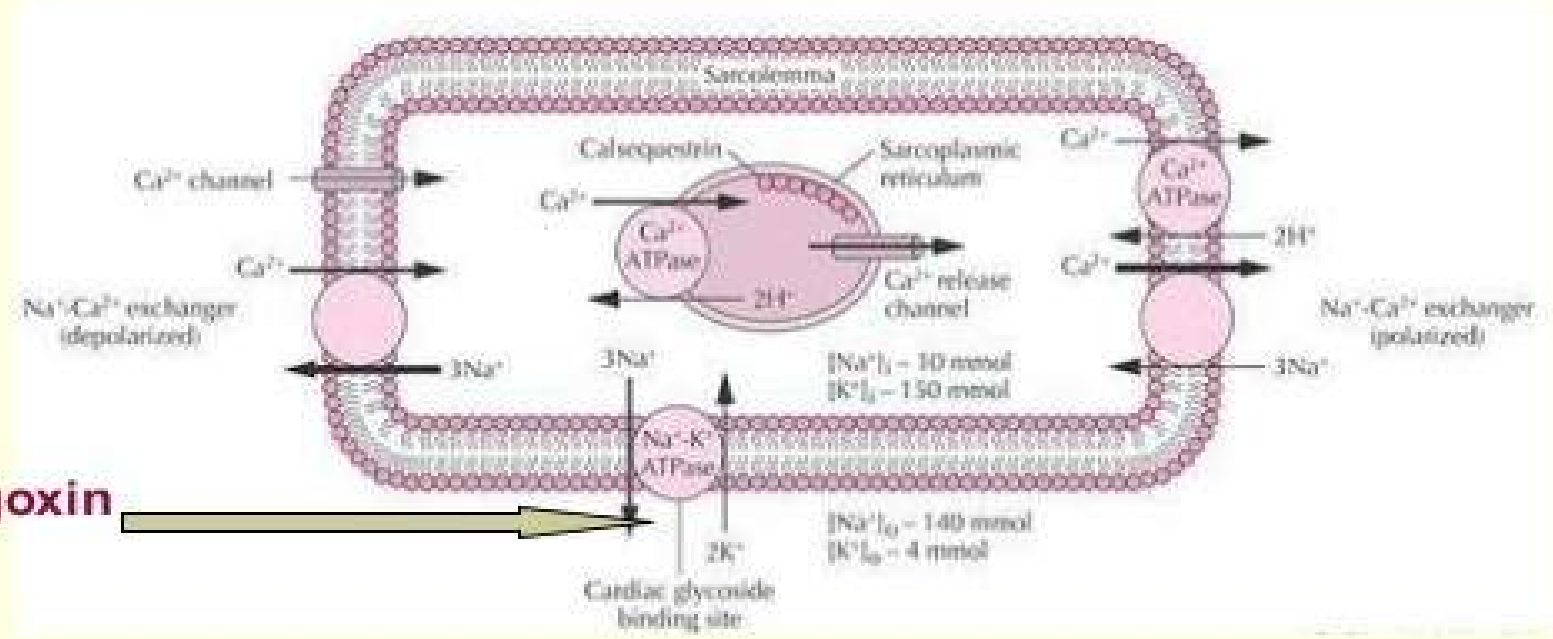
Agar is a gelatinous substance derived from a polysaccharide that accumulates in the cell walls of agarophyte red algae primarily from the genera Gelidium and Gracilaria, or seaweed (Sphaerococcus euchema).

For commercial purpose, it can be obtained from Gelidium amansii and Gelidium cartilaginum.

synonym	Biological source& Family	Chemical constituents	uses	Substituent and adulterants
•Agar-Agar •Vegetable gelatin. 	***Gelidium amansii.*** Family- **Gelidaceae**	**Agarose-**gel strength of agar. **Agaropectin-** viscosity of agar solution.  **Agarose**	•Laxative •Good emulsifying agent •Preparation of jellies •Preparation of Suppositorie s& pessaries	•Danish agar •Indian agar

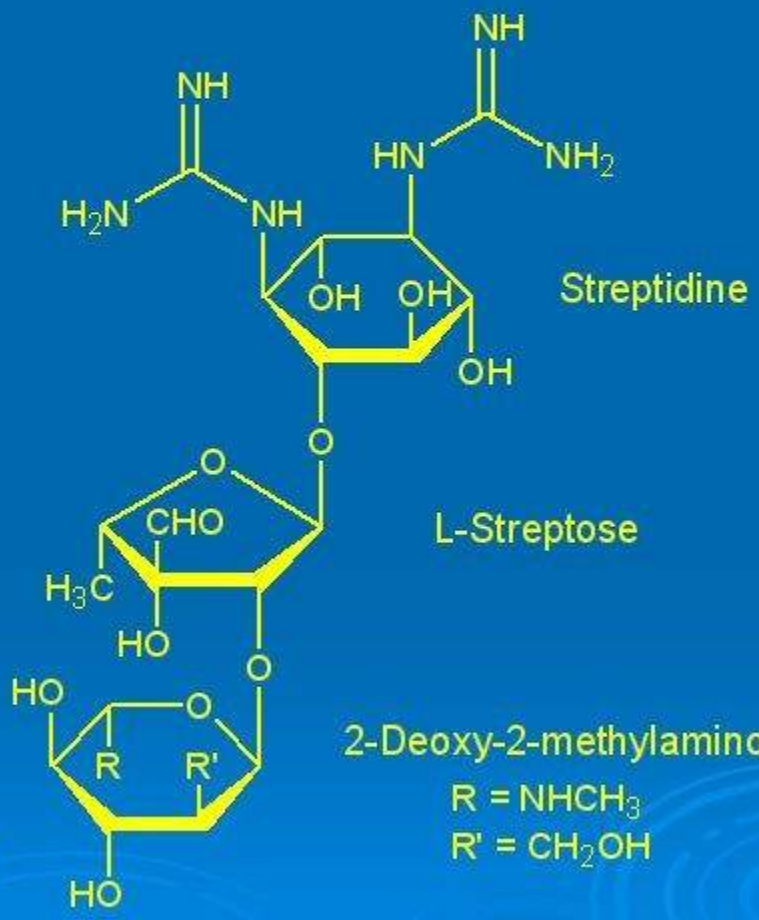
Sodium pump inhibition by cardiac glycosides

- The mechanism by which the cardiac glycosides induce a positive inotropic effect in cardiac muscle is based on the specificity of these drugs for **Na^+K^+-ATPase (the "sodium pump")**



Carbohydrate Antibiotics

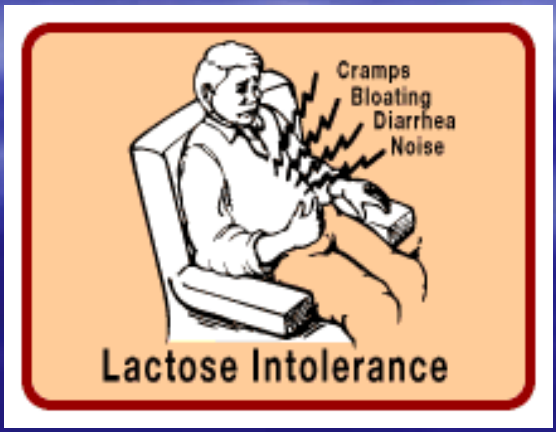
- One of the important discoveries in carbohydrate chemistry was the isolation of the carbohydrate antibiotic called ***streptomycin***.
- Streptomycin is made up of three unusual components.
- The glycosidic linkage is nearly always α .



Streptomycin

Lactose Intolerance

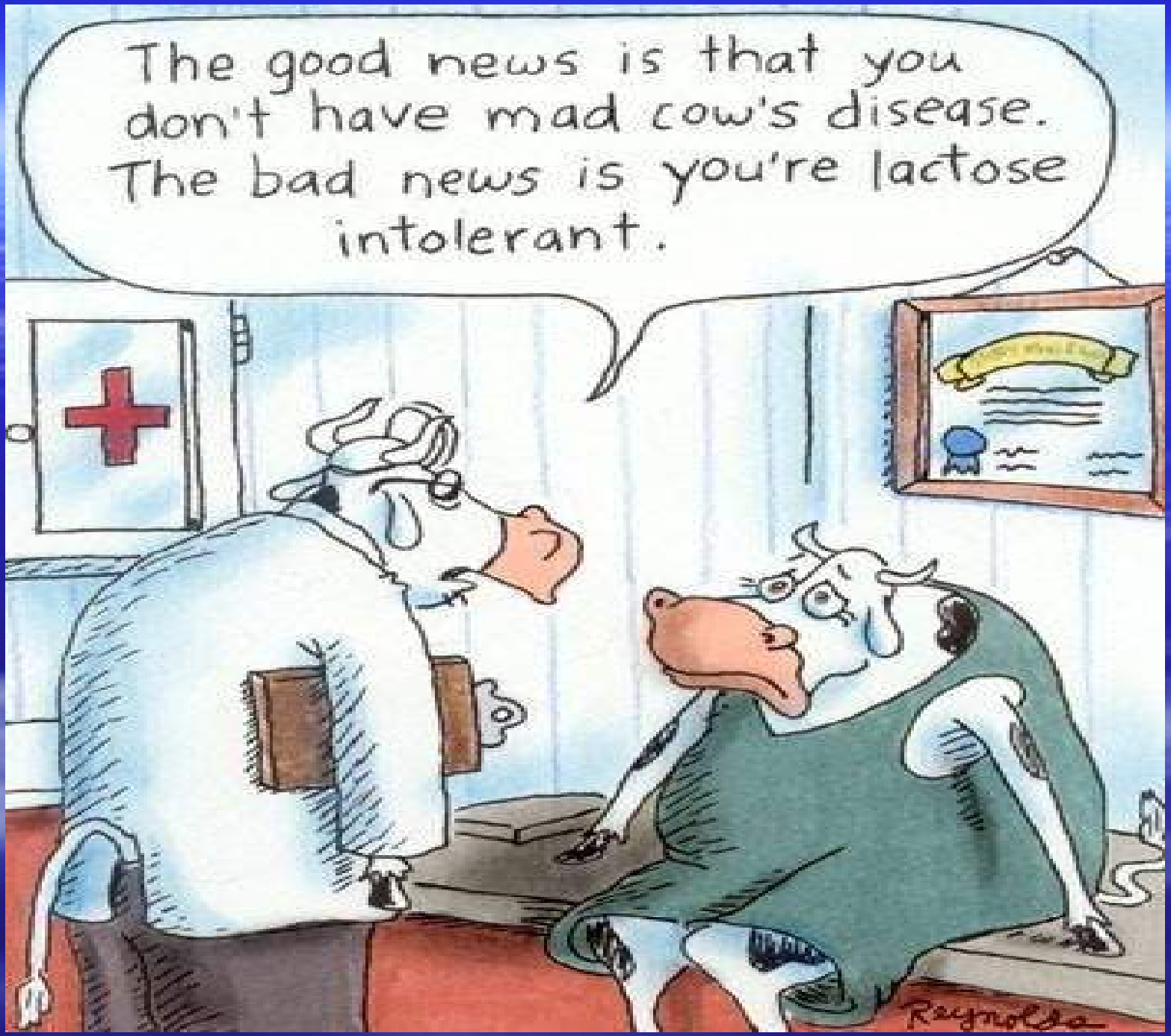
- more lactose is consumed than can be digested
  - lactose molecules attract water
    - cause floating, abdominal discomfort, diarrhea
  - intestinal bacteria feed on undigested lactose
    - produce acid and gas



Lactose Intolerance



- age, damage, medication, diarrhea, malnutrition
- management requires dietary change
  - 6 grams (1/2 cup) usually tolerable
  - take in gradually
  - hard cheeses & cottage cheese
  - enzyme drops or tablets
- lactose free diet is extremely difficult to accomplish


The good news is that you don't have mad cow's disease. The bad news is you're lactose intolerant.
Reynolds

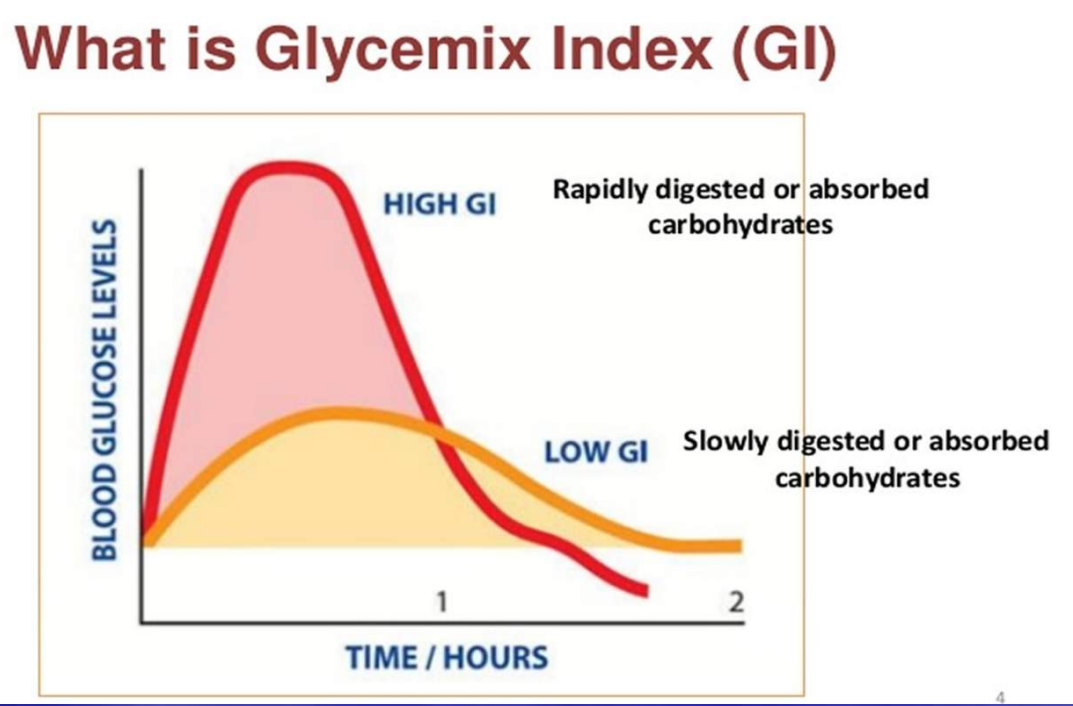
DEXTROSE 5%

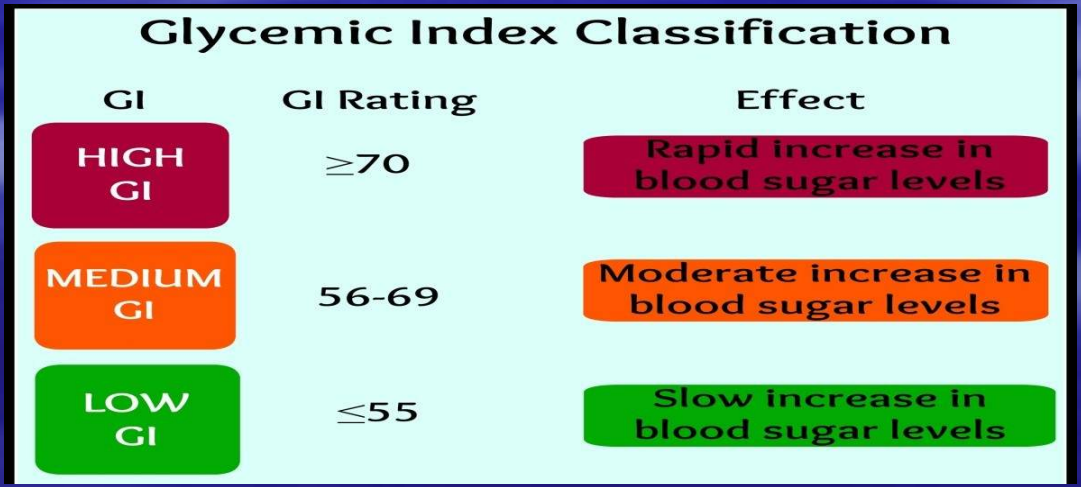
- *Isotonic solution - No electrolytes*
- *50 g/L of glucose*
- *Provide modest calories (1L - 200kcal)*
- *Rapidly metabolized & distribute evenly throughout the all compartments*



Some Important Terms

- ***Glycemic index*** = the blood glucose response to a given food compared to a standard
- ***Glycemic load*** = the amount of Carbohydrate in food times the glycemic index for that food. Related to structure, fiber content, amount of processing, and macronutrient content





Glycemic Index

- way of classifying food according to their ability to raise blood glucose
- much controversy

Glycemic Index of commonly consumed foods	
Based on Glucose as a level of 100	
55-100+ is considered high	
Maltose (as used in beer)	105
Glucose	100
White bread	95
Baked potatoes	95
Carrots (cooked)	85
carrots (juiced)	80
Rice cakes	80
Honey	75
Refined sugar	75
Corn (cooked)	75
Puffed Wheat	75
Cornflakes	75
White rice	70
Shredded wheat	70
Millet	70
Raisins (seedless)	65
Pasta	65
Bananas	60
Couscous	60
Basmati Rice	60
Spaghetti, white	60
Rye Sourdough	55
Wild rice	55
Brown rice	55
Popcorn	55
Kiwi, Grape, Mango	50
Whole-grain pasta	45
Plum, Apple, Orange	40
Lentils & Peas	40
Chick Peas	35
Apricot, dried	30
Milk	30
Nuts	15-30
Hempseed	15
Tomatoes	15
Soy beans	15
Green Vegetables	0-15

Sugar



- ½ comes from natural sources, ½ from refined and added
  - sucrose, corn syrup, honey
- excess can lead to nutrient deficiencies and tooth decay
  - empty calories
  - sugar and starch break down in the mouth

TABLE 4-2 Fibers: Their Characteristics, Food Sources, and Health Effects in the Body

Fiber Characteristics	Major Food Sources	Actions in the Body	Health Benefits
Viscous, soluble, more fermentable • Gums and mucilages • Pectins • Psyllium[a] • Some hemicelluloses	Whole-grain products (barley, oats, oat bran, rye), fruits (apples, citrus), legumes, seeds and husks, vegetables; also extracted and used as food additives.	• Lower blood cholesterol by binding bile. • Slow glucose absorption. • Slow transit of food through upper GI tract. • Hold moisture in stools, softening them. • Yield small fat molecules after fermentation that the colon can use for energy.	• Lower risk of heart disease. • Lower risk of diabetes.
Nonviscous, insoluble, less fermentable • Cellulose • Lignins • Psyllium[a] • Resistant starch • Many hemicelluloses	Brown rice, fruits, legumes, seeds, vegetables (cabbage, carrots, brussels sprouts), wheat bran, whole grains; also extracted and used as food additives.	• Increase fecal weight and speed fecal passage through colon. • Provide bulk and feelings of fullness.	• Alleviate constipation. • Lower risks of diverticulosis, hemorrhoids, and appendicitis. • May help with weight management.

[a] Psyllium, a fiber laxative and cereal additive, has both soluble and insoluble properties.

TABLE 4-3 Fiber in Selected Foods

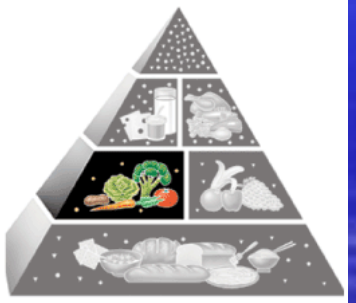
Bread, Cereal, Rice, and Pasta Group



Whole-grain products provide about 1 to 2 grams (or more) of fiber per serving:

- 1 slice whole-wheat, pumpernickel, rye bread.
- 1 oz ready-to-eat cereal (100% bran cereals contain 10 grams or more).
- ½ c cooked barley, bulgur, grits, oatmeal.

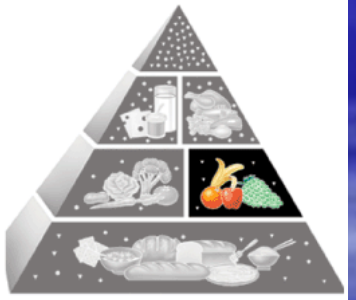
Vegetable Group



Most vegetables contain about 2 to 3 grams of fiber per serving:

- 1 c raw bean sprouts.
- ½ c cooked broccoli, brussels sprouts, cabbage, carrots, cauliflower, collards, corn, eggplant, green beans, green peas, kale, mushrooms, okra, parsnips, potatoes, pumpkin, spinach, sweet potatoes, swiss chard, winter squash.
- ½ c chopped raw carrots, peppers.

Fruit Group



Fresh, frozen, and dried fruits have about 2 grams of fiber per serving:

- 1 medium apple, banana, kiwi, nectarine, orange, pear.
- ½ c applesauce, blackberries, blueberries, raspberries, strawberries.
- Fruit juices contain very little fiber.

Legumes



Many legumes provide about 6 to 8 grams of fiber per serving:

- ½ c cooked baked beans, black beans, black-eyed peas, kidney beans, navy beans, pinto beans.

Some legumes provide about 5 grams of fiber per serving:

- ½ c cooked garbanzo beans, great northern beans, lentils, lima beans, split peas.

NOTE: Appendix H provides fiber grams for over 2000 foods.

Artificial Sweeteners

- help keep sugar and energy intake down
- anything we eat has FDA approval
  - saccharin
  - aspartame
  - acesulfame potassium
  - sucralose
  - neotame

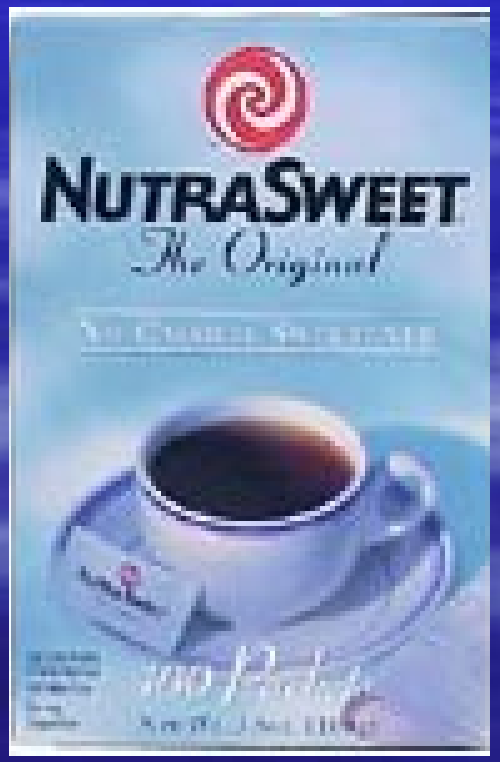


TABLE H4-1 Sweeteners

Sweeteners	Relative Sweetness[a]	Energy (kcal/g)	Acceptable Daily Intake	Average Amount to Replace 1 tsp Sugar	Approved Uses
Approved Sweeteners					
Saccharin	450	0	5 mg/kg body weight	12 mg	Tabletop sweeteners, wide range of foods, beverages, cosmetics, and pharmaceutical products
Aspartame	200	4[b]	50 mg/kg body weight[c] Warning to people with PKU: Contains phenylalanine	18 mg	General purpose sweetener in all foods and beverages
Acesulfame-K	200	0	15 mg/kg body weight[d]	25 mg	Tabletop sweeteners, puddings, gelatins, chewing gum, candies, baked goods, desserts, alcoholic beverages
Sucralose	600	0	5 mg/kg body weight	6 mg	Carbonated beverages, dairy products, baked goods, coffee and tea, fruit spreads, syrups, tabletop sweeteners, chewing gum, frozen desserts, salad dressing
Neotame	8000	0	18 mg/day	0.5µg	Baked goods, nonalcoholic beverages, chewing gum, candies, frostings, frozen desserts, gelatins, puddings, jams and jellies, syrups
Tagatose	0.8	1.5	7.5 g/day	1 tsp	Baked goods, beverages, cereals, chewing gum, confections, dairy products, dietary supplements, health bars, tabletop sweetener
Sweeteners with Approval Pending					**Proposed Uses**
Alitame	2000	4[e]	—		Beverages, baked goods, tabletop sweeteners, frozen desserts
Cyclamate	30	0	—		Tabletop sweeteners, baked goods

[a] Relative sweetness is determined by comparing the approximate sweetness of a sugar substitute with the sweetness of pure sucrose, which has been defined as 1.0. Chemical structure, temperature, acidity, and other flavors of the foods in which the substance occurs all influence relative sweetness.

[b] Aspartame provides 4 kcalories per gram, as does protein, but because so little is used, its energy contribution is negligible. In powdered form it is sometimes mixed with lactose, however, so a 1-gram packet may provide 4 kcalories.

[c] Recommendations from the World Health Organization and in Europe and Canada limit aspartame intake to 40 milligrams per kilogram of body weight.

[d] Recommendations from the World Health Organization limit acesulfame-K intake to 9 milligrams per kilogram of body weight.

[e] Alitame provides 4 kcalories per gram, as does protein, but because so little is used, its energy contribution is negligible.

Sugar Replacers

- sugar alcohols
  - provide bulk and sweetness
    - cookies, gum, candy, jelly
  - do contain minimal kcal
  - low glycemic response
    - absorbed slowly
  - do not cause dental caries





THANK YOU